

Semester - II**THEORY**

Sl. No.	Paper Code	Paper Name	Contact Hours / Week				Credit
			L	T	P	Total	
1	MCAN-201	Data Structure through Python	3	1	-	4	4
2	MCAN-202	Modern Operating System	3	1	-	4	4
3	MCAN-203	Object Oriented Programming with JAVA	3	1	-	4	4
4	MCAN-204	Networking and TCP Suite	3	1	-	4	4
5	Mathematical Elective		3	1	-	4	3
	MCAN-E205A	Numerical and Statistical Analysis					
	MCAN-E205B	Operation Research and Optimization Techniques					

PRACTICAL

1	MCAN-291	Data Structure Lab using Python	-	-	4	4	3
2	MCAN-292	Operating System Lab (Unix)	-	-	4	4	3
3	MCAN-293	Object Oriented Programming Lab using JAVA	-	-	4	4	3

SESSIONAL

1	MCAN-281	Web Technology	-	-	4	4	2
Total Weekly Contact Hours and Credit						36	30

BRIDGE COURSE

[Only for Students of Category “B” stated in the “Eligibility” Section]

A minimum 8-week Online Course on Fundamentals of ‘Software Engineering’ or ‘Systems Analysis and Design’ or ‘Business Systems Applications’ or so

Code: MCAN-201		Paper: Data Structure through Python	
Contacts Hours / Week: 4		Total Contact Hours 40	Credit: 4
Course Outcome: After successful completion of this course, students will be able to: ✓ Understand the concept of abstract data type such as stack, queue, linked list, and trees ✓ Chose appropriate data structure to design algorithm to solve the problem. ✓ Analyze the algorithms in the context of efficiency. ✓ Apply the knowledge of stack and queue to design algorithm ✓ Design application using sorting, searching and the concept of tree.			
UNITS	COURSE CONTENT		
1	Introduction (6L) Basic Terminologies: Elementary Data Organizations, Data Structure Operations: insertion, deletion, traversal etc.; Analysis of an Algorithm, Asymptotic Notations, Time-Space trade off. Searching: Linear Search and Binary Search Techniques and their complexity		
2	Stacks and Queues (8L) ADT Stack and its operations: Algorithms and their complexity analysis, Applications of Stacks: Expression Conversion and evaluation – corresponding algorithms and complexity analysis. ADT, queue, Types of Queue: Simple Queue, Circular Queue, Priority Queue; Operations on each type of Queue: Algorithms and their analysis.		
3	Linked Lists (6L) Singly linked lists: Representation in memory, Algorithms of several operations: Traversing, Searching, Insertion into, Deletion from linked list; Linked representation of Stack and Queue, Header nodes, Doubly linked list: operations on it and algorithmic analysis; Circular Linked Lists: all operations their algorithms and the complexity analysis.		
4	Trees (6L) Basic Tree Terminologies, Different types of Trees: Binary Tree, Threaded Binary Tree, Binary Search Tree, AVL Tree; Tree operations on each of the trees and their algorithms with complexity analysis. Applications of Binary Trees. B Tree, B+ Tree: definitions, algorithms and analysis.		
5	Graph (6L) Graph Terminology, Representation of graphs, Path Matrix, Graph Traversal, BFS, DFS, Minimum Spanning Tree, Kruskal’s Algorithm and Prim’s Algorithm.		
6	Sorting (8L) Objective and properties of different sorting algorithms: Selection Sort, Bubble Sort, Insertion Sort, Quick Sort, Merge Sort, Heap Sort; Performance and Comparison among all the methods Hashing: Review of Hashing, Hash Function, Collision Resolution Techniques in Hashing, Separate Chaining, Open Addressing, Linear Probing, Quadratic Probing, Double Hashing, Rehashing, Extendible Hashing.		
Reference Books: • Fundamentals of Data Structures of C, Ellis Horowitz, Sartaj Sahni, Susan Anderson-freed. • Data Structures in C, Aaron M. Tenenbaum. • Data Structures, S. Lipschutz. • Introduction to Algorithms, Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, lifford Stein.			

Code: MCAN-202**Paper: Modern Operating System****Contacts Hours / Week: 4****Total Contact Hours: 40****Credit:****Course Outcome:**

After successful completion of this course, students will be able to:

- ✓ Describe the main components of OS and their working
- ✓ Explain the concepts of process and thread and their scheduling policies
- ✓ Explain the various memory management techniques
- ✓ Compare the different techniques for managing memory, I/O, disk and files
- ✓ Explains the security and protection features of an Operating System

UNITS	COURSE CONTENT
1	Introduction (8L) Generations Concept of Operating systems, Systems, Types of Operating Systems, OS Services, System Calls, Structure of an OS - Layered, Monolithic, Microkernel Operating Systems, Concept of Virtual Machine. Real Time Operating Systems, Distributed Operating Systems, Multiprocessor Operating System.
2	Process and Scheduling (12L) Processes and Threads: 7 state process model, Process scheduling, Operations on processes, Inter-process communication, Threads overview, Benefits of threads, User and kernel threads. CPU Scheduling: Scheduling criteria, Preemptive & non-preemptive scheduling, Scheduling algorithms (FCFS, SJF, RR, Priority, Multi-level queue, Multi-level feedback queue), Comparative study of the algorithms, Multi-processor scheduling. Process Synchronization: Background, Critical section problem, Software solution – Peterson and Bakery algorithm, Synchronization hardware, Semaphores, Classical problems of synchronization. Deadlocks: System model, Deadlock characterization, Methods for handling Deadlocks, Deadlock prevention, Deadlock avoidance, Deadlock detection, Recovery from deadlock. Case Study: Scheduling on Unix and Windows Operating Systems
3	Memory Management (10L) Basic concept, Logical and Physical address map, Memory allocation: Contiguous Memory allocation– Fixed and variable partition– Internal and External fragmentation and Compaction; Paging: Principle of operation –Page allocation Hardware support for paging, Protection and sharing, Disadvantages of paging. Virtual Memory: Basics of Virtual Memory – Hardware and control structures – Locality of reference, Page fault, Working Set , Dirty page/Dirty bit – Demand paging, Page Replacement algorithms(Optimal, FIFO, SC, NRU and LRU), Thrashing Case Study: Unix Virtual Memory, Windows Virtual Memory
4	File Systems and I/O Management (7L) File concept, Fundamental File System Organization and Access Methods, Directory structure, File system structure, Allocation methods (Contiguous, Linked, Indexed), Free-space management (Bit vector, Linked list, Grouping), Directory Implementation (Linear list, Hash table), Efficiency and Performance. PC Bus Structure, I/O connections, Data transfer techniques (Programmed, Interrupt driven, DMA), Bus arbitration (Daisy chain, Polling, Independent request), Blocking and non-blocking I/O, Kernel I/O subsystem (Scheduling, Buffering, Caching, Spooling and device reservation, Error handling). Case Study: Unix File System, Windows File System
5	Security and Protection: (3L) Overview of Security and Protection, Goals of Security and Protection, Security Attacks, Formal and Practical aspects of Security, Encryption, Authentication and Password Security, Access Descriptors and the Access Control Matrix, Protection Structures, Capabilities, Case Study: Unix Security, Windows Security.

Reference Books:

- Operating System Concepts Essentials, 10th Edition by AviSilberschatz, Peter Galvin, Greg Gagne, Wiley Asia Student Edition.
- Operating Systems: Internals and Design Principles, 5th Edition, William Stallings, Prentice Hall of India.
- Operating System Concepts, EktaWalia, Khanna Publishing House (AICTE Recommended Textbook – 2018)
- Operating System: A Design-oriented Approach, 1st Edition by Charles Crowley, Irwin Publishing
- Operating Systems: A Modern Perspective, 2nd Edition by Gary J. Nutt, Addison- Wesley
- Design of the Unix Operating Systems, 8th Edition by Maurice Bach, Prentice-Hall of India
- Understanding the Linux Kernel, 3rd Edition, Daniel P. Bovet, Marco Cesati, O'Reilly and Associates

Code: MCAN-203 **Paper: Object Oriented Programming with Java**
Contacts Hours / Week: 4 **Total Contact Hours: 40** **Credit: 4**

Course Outcome:

After successful completion of this course, students will be able to:

- ✓ Use the characteristics of Java language in a program. Use variables and data types in program development.
- ✓ Identify and implement arrays, String and Selection Statements.
- ✓ Write Java programs using object-oriented programming techniques including classes, objects, methods, instance variables, interface.
- ✓ Design and implementation programs of Exception handling, Packages, Multithreading Programming, Window based programs.

UNITS	COURSE CONTENT
1	Object-Oriented Languages (10L) Java's History, Creation of Java, Internet & Java, Byte-code, Its Features, Java Program Structure and Java's Class Library, Data Types, Variables, and Operators, Operator Precedence; Selection Statements, Scope of Variable, Iterative Statement; Defining Classes & Methods, Creating Objects of a Class, Defining and Using a Class, Automatic Garbage Collection. Arrays and Strings: Arrays, Arrays of Characters, String Handling Using String Class, Operations on String Handling Using, String Buffer Class.
2	Classes and Inheritance (10L) Using Existing Classes, Class Inheritance, Choosing Base Class, Multiple Levels of Inheritance, Abstraction through Abstract Classes, Using Final Modifier, Packages: Understanding Packages, Defining a Package, Packaging up Your Classes, Adding Classes from a Package to Your Program, Understanding CLASSPATH, Standard Packages, Access Protection in Packages, Concept of Interface. Exception Handling: The concept of Exceptions, Types of Exceptions, Dealing with Exceptions, Exception Objects, Defining Your Own Exceptions.
3	Multithreading Programming (10L) The Java Thread Model, Understanding Threads, The Main Thread, creating a Thread, Creating Multiple Threads, Thread Priorities, Synchronization. Input / Output in Java: I/O Basic, Byte and Character Structures, I/O Classes, Reading Console Input, Writing Console Output, Reading and Writing on Files, Random Access Files, Storing and Retrieving Objects from File, Stream Benefits. Creating Applets in Java: Applet Basics, Applet Architecture, Applet Life Cycle, Simple Applet Display Methods, The HTML Applet Tag Passing Parameters to Applets.
4	Working with Windows (10L) AWT Classes, Window Fundamentals, Working with Frame, Creating a Frame Window in an Applet; Displaying Information within a Window. Working with Graphics and Texts: Working with Graphics, Working with Color, Setting the Paint Mode, Working with Fonts, Managing Text Output; Using Font Metrics, Exploring Text and Graphics, Working with AWT Controls, Layout Managers and Menus.

Reference Books:

- The Complete Reference JAVA, Herbert Schildt, TMH Publication.
- JAVA and Object-Oriented Programming Paradigm, Debasish Jana, Prentice Hall of India
- Beginning JAVA, Ivor Horton, WROX Publication.
- JAVA 2 UNLEASHED, Tech Media Publications.
- JAVA 2(1.3) API Documentations.

Code: MCAN-204**Paper: Computer : Networking and TCP Suite****Contacts Hours / Week: 4****Total Contact Hours: 40****Credit: 4****Course Outcome:**

After successful completion of this course, students will be able to:

- ✓ Understand the purpose of network layered models, network communication using the layered concept and able to compare and contrast OSI and TCP/IP model.
- ✓ Differentiate among and discuss the four level of address (physical, logical, port and url) used by the internet TCP/IP protocols.
- ✓ Understand the routing principals and algorithm such as distance vector routing and link state.
- ✓ Judge the efficiency of the connection oriented and connectionless protocol.
- ✓ Familiar with the routing techniques, protocols and quality of service.
- ✓ Explain the concept of network security and cryptography.

UNITS	COURSE CONTENT
1	Introduction (4L) Direction of data flow (simplex, half duplex, full duplex), Network topology, categories of network (LAN, MAN, WAN).
2	Protocol and Standard (4L) Layered Task, The OSI model, TCP/IP protocol suite, Addressing
3	Internetworking (10L) Internetworking concept, IPv4 and IPv6 Addressing, IPv4 protocol, IPv6 protocol, transition from IPV4 to IPV6, transition from IPv4 to IPv6, Address Mapping, Error Reporting, Multicasting, Unicast Routing Protocols, Distance Vector routing, Link state routing, Path vector routing, Multicasting Routing Protocols, Transmission Control Protocol(TCP), User Datagram Protocol(UDP)
4	Quality of Service (6L) Data traffic, Congestion, congestion control, Quality of service, Techniques to improve QoS, Integrated services, Differentiated service, QoS in Frame Relay, QoS in ATM
5	DNS and Web (8L) Name Space, Domain Name System, Distribution of Name Space, Remote Logging, Electronic Mail and File Transfer, WWW, Web document and HTTP, Network Management, Simple Network Management Protocol (SNMP)
6	Network Security (8L) Symmetric Key Cryptography, DES, AES, Asymmetric Key Cryptography, RSA, Diffie-Hellman, Security Services, Digital Signature, Key Management, IP Security, SSL/TLS, PGP, Firewalls

Reference Books:

- Computer Networks, Andrew S. Tanenbaum, Pearson Education, Fourth edition.
- Data and Computer Communication, William Stallings, Prentice hall, Seventh edition.
- High speed Networks and Internets, William Stallings, Pearson education, Second edition.
- Behrouz A Forouzan, - Data communication & Networking , TMH
- Behrouz A Forouzan, - TCP/IP Protocol Suite , TMH
- Kelvin R Fall, W. Richard Stevens- TCP/IP Illustrated Volume 1, Addison Wesley

Code: MCAN-E205A		Paper: Numerical and Statistical Analysis	
Contacts Hours / Week: 4		Total Contact Hours: 40	Credit: 3
Course Outcome: After successful completion of this course, students will be able to: ✓ To understand approximation and propagation error. ✓ To understand and implement different interpolation techniques. ✓ To understand and implement integration techniques. ✓ To understand and implement solutions for linear and algebraic and differential equations.			
UNITS	COURSE CONTENT		
1	Approximation in numerical computation (3L) Truncation and rounding errors, Fixed and floating point arithmetic, Propagation of errors.		
2	Interpolation (5L) Newton forward/backward interpolation, Lagrange’s and Newton’s divided difference Interpolation		
3	Numerical integration (4L) Trapezoidal rule, Simpson’s1/3 rule, Romberg’s Integration, Expression for corresponding error terms.		
4	Numerical solution of Linear equations (6L) Gauss elimination method, Matrix inversion, LU Factorization method, Gauss-Seidel iterative method.		
5	Numerical solution of Algebraic equation (5L) Bisection method, Regula-Falsi method, Newton-Raphson method, Iteration Method, Secant Method.		
6	Numerical solution of ordinary differential equation (5L) Euler’s method, Runge-Kutta methods, Taylor’s series, method, Predictor Corrector methods and Finite Difference method.		
7	Least Square Curve fitting (3L) Linear & non-linear curve fitting		
8	Introduction to Statistics& Probability (9L) Basic Statistics-measure of central tendency, dispersion. Probability, distribution introduction to mass function, density function, distribution function (Binomial, Poisson, Normal).		
Reference Books: • R.S. Salaria: Computer Oriented Numerical Methods, Khanna Publishing House • C.Xavier: C Language and Numerical Methods. • Dutta & Jana: Introductory Numerical Analysis. • J.B.Scarborough: Numerical Mathematical Analysis. • Jain, Iyengar, & Jain: Numerical Methods (Problems and Solution). • Balagurusamy: Numerical Methods, Scitech. • Baburam: Numerical Methods, Pearson Education. • N. Dutta: Computer Programming & Numerical Analysis, Universities Press			

Code: MCAN-E205B Paper: Operation Research and Optimization Techniques
Contacts Hours / Week: 4 Total Contact Hours: 40 Credit: 3

Course Outcome:

After successful completion of this course, students will be able to:

- ✓ 1. Describe the way of writing mathematical model for real-world optimization problems.
- ✓ 2. Identify Linear Programming Problems and their solution techniques
- ✓ 3. Categorize Transportation and Assignment problems
- ✓ 4. Apply the way in which Game Theoretic Models can be useful to a variety of real-world scenarios in economics and in other areas.
- ✓ 5. Convert practical situations into non-linear programming problems.
- ✓ 6. Solve unconstrained and constrained programming problems using analytical techniques.

UNITS	COURSE CONTENT
1	Linear Programming Problem (LPP)-I (10L) Formulation of an LPP; Graphical Method of solution of an LPP; Convex Combination and Convex Set; Convex Hull and Convex Polyhedron; Canonical and Standard form of an LPP; Basic Solution of a system of linear equations; Simplex Method; Big-M Method; Concept of Duality; Mathematical formulation of duals.
2	Linear Programming Problem (LPP)-II (10L) Transportation Problems (TP) ; Representation of Transportation Problems as LPP; Methods of finding initial basic feasible solution of TP: North-West Corner Rule, Matrix Minima Method, Vogel's Approximation Method; Optimality test of the basic feasible solution; Assignment Problems; Hungarian Method.
3	Game Theory (10L) Introduction; Strategies; The Minimax and Maximin Criterion; Existence of Saddle Point; Two person zero sum Games; Games with saddle Point – Pure Strategies; Games without a Saddle Point – Mixed Strategies; Symmetric Games; Dominance Principle; Graphical Method of Solution; Algebraic Method of Solution.
4	Non-Linear Programming Problem (NLPP) (10L) Single-variable Optimization; Multivariate Optimization with no constraints: Semidefinite Case, Saddle Point; Multivariate Optimization with Equality Constraints: Method of Lagrange Multipliers; Multivariable Optimization with inequality constraints: Kuhn-Tucker Conditions.

Reference Books:

- Linear Programming and Game Theory by J. G. Chakraborty and P. R. Ghosh, Moulik Library.
- Operations Research by Kanti Swarup, P. K. Gupta and Man Mohan, S. Chand and Sons.
- Engineering Optimization by S. S. Rao, New Age Techno Press.
- Operations Research by J K Sharma, Macmillan India Ltd

Code: MCAN-291**Paper: Data Structure Lab using Python****Contacts Hours / Week: 4****Total Contact Hours: 40****Credit: 3****Course Outcome:**

After successful completion of this course, students will be able to:

- ✓ To understand linear and non-linear data structures.
- ✓ To understand different types of sorting and searching techniques.
- ✓ To know how to create an application specific data structure.
- ✓ To solve the faults / errors that may appear due to wrong choice of data structure.
- ✓ To analyze reliability of different data structures in solving different problems.

UNITS	COURSE CONTENT
1	Implementation of data structure operations (Insertion, deletion, traversing, searching) on array. Linear search, Binary search.
2	Implementation of stack, queue operation using array. Pop, Push, Insertion, deletion, Implementation of circular queue. Infix to postfix conversion, postfix expression evaluation
3	Implementation of linked lists: Single linked list, circular linked list, double linked list, doubly circular linked list. Implementation of stack and queue using linked list. Merging two linked list, Linked list representation of a polynomial, polynomial addition, polynomial multiplication.
4	Tree: creating Binary Search tree, recursive and non-recursive traversal of BST, deletion in BST, calculating height of a BST, building AVL tree.
5	Implementation of sorting techniques: selection, bubble, quick sort, insertion sort, merge sort, heap sort, implementation of priority queue. Hash table implementation.
6	Implementation of Graph: representation, searching, BFS, DFS

Code: MCAN-292		Paper: Operating System Lab (Unix)	
Contacts Hours / Week: 4		Total Contact Hours: 40	Credit: 3
Course Outcome: After successful completion of this course, students will be able to: ✓ Do the use of basic UNIX Commands from the command line, and create Shell Scripts to customize their UNIX Working Environment. ✓ Organize and manage their processes within UNIX through system calls. ✓ Organize and manage their files within the UNIX through system calls. ✓ Provide a mechanism for handling asynchronous events through signals (Software Interrupt). ✓ Implement the Inter-process communication using FIFOs, Message Queues, Semaphores, and Shared Memory. ✓ Explain Socket programming to design Client-Server Environment. ✓ Understand and implement Multithreaded Programming Environment.			
UNITS	COURSE CONTENT		
1	Shell programming Creating a script, making a script executable, shell syntax (variables, Conditions, control structures, functions and commands).		
2	Process Starting new process, replacing a process image, duplicating a process image, waiting for a process, Zombie Process, Orphan Process		
3	File Handling Programming on files (use create(), open(), read(), write(), close(), lseek(), dup()).		
4	Signal Signal Handling, Blocking, Suspending, Delivering Signals, Various Signal Related Functions.		
5	Inter-process communication Pipes (use functions pipe(), popen(), pclose()), Named Pipes (FIFOs, accessing FIFO), Message Queues (use functions msgget(), msgsnd(), msgrcv(), msgctl()), Semaphores (use functions semctl(), semget(), semop()) Shared Memory (use functions shmget(), shmat(), shmdt(), shmctl())		
6	Sockets: TCP Sockets, UDP Sockets, Socket Options, Client /Server Example, Name and Address Conversions		
7	POSIX Threads Programming with pthread functions (viz. pthread_create(), pthread_join(), pthread_exit(), pthread_attr init(), pthread_cancel())		

Code: MCAN-293 **Paper: Object Oriented Programming Lab using JAVA**
Contacts Hours / Week: 4 **Total Contact Hours: 40** **Credit: 3**

Course Outcome:

After successful completion of this course, students will be able to:

- ✓ Apply object-oriented principles or features in software design process to develop Java programs for real life applications.
- ✓ Reduce the complexity of procedural language by employing different OOP technologies for developing robust and reusable software.
- ✓ Develop programs using stream classes for various I/O operations and design concurrent.
- ✓ Design graphical user interface to develop user interactive applications.

UNITS	COURSE CONTENT
1	Assignments on class, constructor, overloading, inheritance, overriding.
2	Assignments on wrapper class, arrays.
3	Assignments on developing interfaces- multiple inheritance, extending interfaces.
4	Assignments on creating and accessing packages.
5	Assignments on multithreaded programming
6	Assignments on applet programming

Code: MCAN-281		Paper: Web Technology	
Contacts Hours / Week: 4		Total Contact Hours: 40	
Credit: 2			
Course Outcome: After successful completion of this course, students will be able to: ✓ Understand the basic working methodology of HTML, CSS, java script ✓ Apply In-Built and Create User defined functions of java script for form validation.. ✓ Students are able to develop a dynamic webpage by the use of java script and DHTML.. ✓ Debug the Programs by applying concepts and error handling techniques.			
UNITS		COURSE CONTENT	
1	Understanding of Internet www, client-server, DNS, IP Protocol, HTTP, URL, Browser working principal, Web Hosting, W3C standard, Cloud Development.		
2	Fundamental of Web Design HTML: Introduction, Editor(VS Code/ Sublime), Element, Attribute, Head, Heading, Paragraph, Style, Formatting, Quotation, Comment, Color, CSS, Link, Image, Table, List, Block & Inline, Class, ID, Iframe, Script, File path, Layout, Code, Entity, Symbol, Emoji, Charset, Forms, Form Attributes, Elements, Input types, Input Attributes. CSS: Introduction, Selector, External-Internal-Inline CSS, Comments, Color, Background, Border, Margin, Padding, Height, Width, Box model, Outline, Text, Font, Icon, Link, List, Table, Display, Max width, Position, Overflow, Float, Inline-bock, Align, Pseudo-class, Pseudo-element, Opacity, Navigation Bar, Dropdowns, Image gallery, Image sprites, Attr Selector, Form, Counter, Units, Rounded corner, Border image, Gradient, Shadow, Text Effect, Web Fonts, Transition, Animation, Tooltip, Style Image, Button, Pagination, Multiple column, Media Query, Flexbox.		
3	Advance Web Design CSS Responsive Design: Introduction, Viewport, Grid view, Media queries, Responsive image, Responsive video. Bootstrap: Introduction, Container, Grid, Typography, Color, tables, Images, jumbotron, Alerts, Button, Button group, Badges, Progress bar, Spinner, Pagination, List group, Card, Dropdown, Collapse, Navs, Navbar, Forms, Input, Input group, Carousel, Modal, Tooltip, Popover, Toast, Scrollspy, Flex, Media object. Java Script: Introduction, output, variables, operator, Datatype, Function, Object, Event, String, String method, Number method, Array, Array method, Array iteration, Date & Date format, Date method, Math, Random, Comparison, Condition, For, While, break, This keyword, Function, Arrow function, Form validation, HTML DOM – Documents, Elements, HTML, CSS, Animation, Event, Even listener, Navigation, Nodes, Collection, Node list.		